

The Viable Selector: A Unifying Framework for Obstruction Certificates under Incomplete Observation

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Abstract

Every obstruction certificate in the calculus of viability under incomplete observation proves the same structural fact: at some information state, no single response acceptable for every compatible branch exists at the required level of the recursion. This paper develops that fact into a framework. The recursive viable-selector correspondence is defined, and the finite-horizon identity is established: an information state is viable exactly when the correspondence at it is nonempty, with the measurable-selection qualification that confines exact completeness claims to finite or suitably regular models. A meta-theorem organizes the certificate families of the companion papers — dynamic exit, admissibility, common action, tube safety, timing, certification limits, and policy-class restrictions — as emptiness conditions on correspondences obtained from one another by nesting, restriction, and label selection, with three bridging propositions making the passages precise: the correspondence ladder, timing as blind-window predecessor emptiness, and certification as label selection. The framework unifies the form of the certificates, not their cause: dynamic exit is not epistemic, the fibre criterion is a certification limit rather than a policy-viability obstruction, and the certainty-equivalence trap is a policy-class restriction that can operate under injective observation. All structural identities are verified exactly on the worked systems of the companion papers.

1. Introduction

Two companion papers develop a calculus of obstruction certificates — finite, checkable witnesses that no observation-based policy is viable. The first (Abaee, 2026a) establishes the certificate families for discrete and continuous systems; the second (Abaee, 2026b) builds the continuous-to-finite bridge for the recourse-aware mode. The mechanisms can look like a taxonomy: an exit condition here, a timing bound there, a fibre criterion, a policy-class trap. The observation of this paper is that they are one theorem worn seven ways. At an information state B , an observation-based regulator must choose *one* response that works for *every* compatible branch; each certificate is a proof that, at some recursion level, no such response exists.

The paper makes this precise and then measures how much it carries.

- **The recursive viable selector** (Section 2): the correspondence $\Gamma_N(B)$ of response segments that are admissible for the declared class, implementable under the observation structure, safe through the current review interval, and recursively viable at every successor information state. Theorem 1: in finite-horizon finite models, B is viable exactly when $\Gamma_N(B) \neq \emptyset$ — and the measurable-selection qualification states exactly where pointwise nonemptiness must be upgraded to an implementable selector.
- **The meta-theorem** (Section 3, Theorem 2): declaring a policy class intersects the selector with the class's requirements; an obstruction certificate *is* a checkable condition implying the corresponding emptiness; and soundness is monotone — restricting the class or tightening the requirements preserves every verdict. The certainty-equivalence trap is the canonical instance: the unrestricted correspondence is nonempty while the class-restricted one is empty.
- **Three bridging propositions** (Section 4): the correspondence ladder $\mathcal{A}_N(B) \subseteq \mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_V^B(B) \subseteq U^B(B)$, under precise hypotheses, so that emptiness at any coarser link certifies recursive nonviability; the identity of the guaranteed survival time σ^* with the horizon at which the blind-window predecessor correspondence empties, so the timing certificate *is* predecessor emptiness; and the label-selection form of the fibre criterion, $\Lambda_K(O^{-1}(y)) \neq \emptyset$ for every y , with the certainly-safe set as the maximal sound label relaxation.
- **The specializations** (Section 5): the certificate families of the companions derived as instances, in one table, with the correspondence emptied and the certificate object named in each row.
- **Verified re-derivations** (Section 6): the two-floor belief, the two-patch audit, the hidden-regime grid, and the certainty-equivalence instance of the companion papers, re-derived as framework identities in exact arithmetic (seven check families, all passing).
- **The discipline** (Section 7): what the framework does *not* unify. The form is common; the cause is not. Dynamic exit defeats full information and is not epistemic; the fibre criterion limits certification, not policy viability; the certainty-equivalence trap is a restriction of the policy class and can operate under injective observation.

The strongest claim the framework supports is stated at the end of Section 7 and is deliberately modest: observation-based viability requires a common recursively safe selector from information states to admissible responses; the certificates are checkable sufficient emptiness conditions; the backward recursion characterizes selector existence exactly in finite-horizon finite models; elsewhere the certificates are sound local exclusion tests.

2. The recursive viable selector

2.1 States, information, responses

Let Z be the augmented state space: a point $z \in Z$ may carry the physical state, hidden parameters, latent regimes, or institutional modes affecting admissibility. Let $\mathcal{V} \subseteq Z$ be the closed safe set. An *information state* is a nonempty set $B \subseteq Z$ of augmented states compatible with the observation record. At known times $0 = \tau_0 < \tau_1 < \dots < \tau_R$, the observation history identifies a cell of a refining partition of the compatibility relation; within a review interval I , a *response segment* α is a held action, an open-loop segment, or an observation-based feedback law, as the model admits. For a response α at B and a next observation y^+ , $\text{Succ}(B, \alpha, y^+)$ denotes the successor information state. The exposition is written for the deterministic set-valued semantics of the companions; the stochastic extension is the belief-state value recursion previewed in Section 8.

2.2 The correspondence

Fix a policy class Π , an observation structure $\mathcal{I}$, a review length Δ , and a horizon N .

Definition 1 (recursive viable selector). $\mathfrak{R}_{\Pi, \mathcal{I}}(B)$ is the set of response segments that belong to the class Π , are implementable under $\mathcal{I}$, and satisfy the nonanticipativity convention. Inductively, $\mathscr{W}_0$ is the collection of information states carried by $\mathcal{V}$, and

$$\Gamma_N(B) = \left\{ \alpha \in \mathfrak{R}_{\Pi, \mathcal{I}}(B) : \text{every compatible trajectory remains in } \mathcal{V} \text{ for } N \text{ steps} \right\}$$

$$\mathscr{W}_N = \{ B : \Gamma_N(B) \neq \emptyset \}.$$

Theorem 1 (finite-horizon exactness). In a finite-horizon finite model, $B \in \mathscr{W}_N$ if and only if $\Gamma_N(B) \neq \emptyset$: an N -step viable policy exists from B exactly when the recursively viable response correspondence at B is nonempty.

Proof. Backward induction on N . For $N = 0$, viability is membership in $\mathcal{V}$, which is the definition of $\mathscr{W}_0$. If B is viable for N steps, the first segment of a witnessing policy is admissible, keeps every compatible trajectory in $\mathcal{V}$ through the interval, and every possible successor information state is viable for $N - 1$ steps, so the segment lies in $\Gamma_N(B)$; conversely, selecting recursively from nonempty Γ_k along the realized observation path assembles a viable policy. $\square$

Remark 1 (the selector qualification). Pointwise nonemptiness of $\Gamma_N(B)$ is enough for backward induction in a finite model. In general measurable spaces a policy requires an implementable *selection* $B \mapsto \alpha(B) \in \Gamma_N(B)$, and the following are distinct: nonemptiness at one specified information state; nonemptiness at every state of a

collection; and the existence of a measurable nonanticipative selector on that collection. Exact completeness claims are therefore made only for finite or suitably regular models (measurable selection under standard conditions; Bertsekas and Shreve, 1978); elsewhere the certificates remain sound exclusion tests, which require no selection.

An *obstruction certificate* is any checkable condition implying $\Gamma_N(B) = \emptyset$. The certificates of the companions are all of this form; the next two sections say precisely which correspondence each empties and by which passage the emptiness is transferred to Theorem 1.

3. The meta-theorem

Declare a policy class by four requirement families: admissibility under the class rules, implementability under the observation structure, tube safety over the review interval at the declared length, and recursive viability at the declared horizon. Write

$$\Gamma(B; \Pi, \mathcal{I}, \Delta, N) = \Gamma^{\text{adm}}(B) \cap \Gamma^{\text{imp}}(B) \cap \Gamma^{\text{tube}}(B; \Delta) \cap \Gamma^{\text{rec}}(B; N),$$

the intersection of the correspondences enforcing each family.

Theorem 2 (meta-theorem). B is viable within the declared class if and only if $\Gamma(B; \Pi, \mathcal{I}, \Delta, N) \neq \emptyset$. Consequently: (i) a checkable condition implying the emptiness of any one factor is an obstruction certificate for the whole class; (ii) the certificate semantics of the companions — a finite, policy-independent list of adversarial tests whose violation is forced — are exactly the dual witnesses of such emptinesses; (iii) soundness is monotone: if $\Pi' \subseteq \Pi$, or if any factor correspondence is tightened, then $\Gamma(B; \Pi', \mathcal{I}, \Delta, N) \subseteq \Gamma(B; \Pi, \mathcal{I}, \Delta, N)$, so every emptiness verdict for the larger class remains valid for the smaller, and new verdicts may appear.

Proof. (i) and the identity are Theorem 1 with the class requirements carried in $\mathfrak{R}_{\Pi, \mathcal{I}}$; (ii) is the certificate semantics of the companions restated (a Farkas vector, an adversarial label list, or a timing bound is a finitely checkable witness of the emptiness of the relevant factor); (iii) is immediate from the intersection form. $\square$

Corollary 1 (policy class and $\text{Succ}(B, \alpha, y^+)$). If $\Gamma(B; \Pi, \mathcal{I}, \Delta, N) \neq \emptyset$ but $\Gamma(B; \Pi', \mathcal{I}, \Delta, N) = \emptyset$ for a restriction $\Pi' \subset \Pi$, then B is viable under Π and nonviable under Π' — the certainty-equivalence trap of the companion paper, where the restriction removes the required selector even under an injective observation.

Table 1 instantiates the theorem: each certificate family of the companions is the emptiness of one factor, or of a correspondence derived from one by the passages of Section .

4. Bridging propositions

Proposition 1 (the correspondence ladder). Let B be an information state and $\Delta \leq$ one review interval. Under the standing hypotheses of the companions (constraint functions of class C^1 ; disturbance correspondences with closed graph; dynamics admitting the comparison argument), the recursively viable, tube-safe, instantaneously safe, and admissible correspondences satisfy

$$\mathcal{A}_N(B) \subseteq \mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_V^B(B) \subseteq U^B(B),$$

where $\mathcal{A}_N(B) = \Gamma_N(B)$ with the singleton action reading. Each inclusion can be strict. Consequently every emptiness certificate at a coarser link is a certificate of recursive nonviability at the finer one.

certificate family	correspondence emptied	certificate object
dynamic exit (base)	Γ^{rec} at a singleton branch (no intersection left)	drift/comparison bound, exit time $\leq q(x_0)/\varepsilon$
admissibility	$\Gamma^{\text{adm}}(B) = \bigcap_z U(z)$	empty common admissible set
common action	$\Gamma^{\text{safe}}(B) = \bigcap_z \mathcal{R}_V(z)$	empty common safe-action set; Farkas or $m+1$
tube safety	$\Gamma^{\text{tube}}(B; \Delta)$	empty tube-safe set at review length Δ
delayed information (timing)	blind-window predecessor correspondence at horizon T_{obs} (Prop. 2)	$\sigma^*(B_0) < T_{\text{obs}}$; LP values and Farkas duals
certification limit	label selector $\Lambda_K(O^{-1}(y))$ (Prop. 3)	a fibre across the safe-set boundary
policy class	$\Gamma(B; \Pi', \cdot)$ for the restricted class (Cor. 1)	emptiness under the restriction

Table 1: The certificate families of the companion papers as emptiness conditions on declared-class correspondences. The last three rows act on correspondences derived by the passages of Propositions 2 and 3 and Corollary 1; the fibre row is a certification limit, not a policy-viability obstruction (Section).

Proof. An action viable for N steps keeps every compatible trajectory in $\mathcal{V}$ over the first interval, hence is tube-safe at Δ ; a tube-safe action is, at every active boundary state of every compatible branch, a direction of nonnegative constraint drift against every admissible disturbance (otherwise a measurable adverse selection exits within the interval), hence instantaneously safe; instantaneous safety requires admissibility at every branch. Strictness is real: the two-patch instance of Section has $\mathcal{A}_{\text{tube}}(\cdot, 2)$ empty where $\mathcal{R}_V^B$ is a singleton. $\square$

Proposition 2 (timing as blind-window predecessor emptiness). For the delayed-information model (no information before T_{obs} , revelation at T_{obs}), let $\text{Pre}_{\text{blind}}(\mathcal{C})$ be the collection of information states admitting a blind response — admissible for every branch, common across indistinguishable branches, within the declared window class — that keeps every branch in $\mathcal{V}$ through the window with every revealed successor in $\mathcal{C}$, and let $\sigma^*(B_0)$ be the guaranteed blind-window survival time. Then

$$\sigma^*(B_0) = \sup\{t : B_0 \text{ survives } \text{Pre}_{\text{blind}} \text{ at horizon } t\}, \quad \sigma^*(B_0) < T_{\text{obs}} \implies B_0 \notin \text{Pre}_{\text{blind}}(\mathcal{C}) \text{ at } T_{\text{obs}}.$$

The timing certificate of the companions is exactly this emptiness, evaluated at the observation horizon; and the declaration of the window class is load-bearing, since different classes empty at different horizons (Section , hidden-regime grid).

Proof. The sup-identity is the definition of σ^* read through Theorem 1: survival to horizon t is precisely nonemptiness of the blind predecessor correspondence at t . If $\sigma^*(B_0) < T_{\text{obs}}$, no blind response keeps every branch safe to the horizon, so the correspondence at T_{obs} is empty whatever post-reveal collection $\mathcal{C}$ is offered. $\square$

Proposition 3 (certification as label selection). For a label space $\{0, 1\}$ and the safety label $\mathbf{1}_K$, define the exact-label correspondence on a set F of augmented states by $\Lambda_K(F) = \{\ell \in \{0, 1\} : \ell = \mathbf{1}_K(z) \forall z \in F\}$. An exact observation-only certifier exists if and only if $\Lambda_K(O^{-1}(y)) \neq \emptyset$ for every observation y ; the certainly-safe set is $\{y : \Lambda_K(O^{-1}(y)) = \{1\}\}$, the maximal sound label relaxation. The fibre criterion is the label-selection analogue of common-action selection at the zeroth level of the recursion; it is a limit on certification, not a policy-viability obstruction (Section).

Proof. A certifier is a map $\ell : O(Z) \rightarrow \{0, 1\}$ with $\ell(y) = \mathbf{1}_K(z)$ for all $z \in O^{-1}(y)$; such a value exists exactly when $\mathbf{1}_K$ is constant on the fibre, which is nonemptiness of $\Lambda_K(O^{-1}(y))$. The certainly-safe set collects the fibres on which the only constant label is the safe one. $\square$

5. The specializations

The four one-step mechanisms of the companion paper are the static intersection object at four choices of acceptable response: admissibility ($\Gamma(z) = U(z)$); instantaneous safety ($\Gamma(z) = \mathcal{R}_V(z)$); tube safety at a review length ($\Gamma(z) = \mathcal{A}_{\text{tube}}(z, \Delta)$),

the held action reading; and the epistemic-emptiness construction, which is the admissibility register evaluated on a merged observation fibre — one constant observation, two admissible-response sets with empty intersection. The dynamic exit certificate is the base case: at a single branch the intersection is trivial, and the drift condition empties the boundary response set outright — no information-theoretic content. The delayed-information obstruction is Proposition 2: the same intersection, demanded through the blind window and handed to the revealed successors. The certification limit is Proposition 3. The certainty-equivalence trap is Corollary 1. One object — a response acceptable for every compatible branch at the required recursion level — seven emptinesses.

6. Verified re-derivations

The framework’s identities are machine-checked in exact rational arithmetic (`paper2_viable_selector_v1_verify.py`, standard library; seven check families, all passing) on the worked systems of the companions. *Static intersection (two-floor belief)*: the common admissible correspondence is $[0, 1]$ — nonempty — while the common safe correspondence is empty: the common-action certificate is exactly $\Gamma^B = \emptyset$. *Label selection (two-floor index and its two-dimensional aggregate)*: the crossing fibre has $\Lambda_K = \emptyset$, the safe fibre $\Lambda_K = \{1\}$, and the certainly-safe readings are exactly the aggregate values $I \geq \frac{7}{5}$ (seven grid cells), matching the companions’ computation. *Ladder (two-patch system)*: $\mathcal{A}_{\text{tube}}(\cdot, 2) \subseteq \mathcal{R}_V^B \subseteq U^B$ holds at every state, is strict at five — at $(1, 2)$ the two-step held-action correspondence is empty while the instantaneous one is $\{1\}$ — and tight at four. *Recursive identity (two-patch audit)*: the Γ_N correspondence recursion and the belief-kernel recursion return identical verdicts on all six audited beliefs (three viable, with witness selectors; three nonviable, each certified by common-action emptiness), and singleton Γ_8 nonemptiness coincides with the full-information kernel. *Timing (hidden-regime grid, all 48 cells)*: the blind-window predecessor correspondence over the declared hold class is nonempty at horizon k exactly when $z_0 \geq 1+k$ — the viability boundary — while the unrestricted sequential class is nonempty exactly when $z_0 \geq 2$ for $k \geq 2$: Proposition 2’s load-bearing declaration, verified. *Policy class (the certainty-equivalence instance)*: the corrected law has drift identically zero on $[1, 2]$, so the unrestricted correspondence is nonempty, while every certainty-equivalence law has drift at least $\frac{21}{100} > 0$ there and exits in finite time: the class-restricted correspondence is empty, Corollary 1 attained.

7. What is not unified

The framework unifies the *form* of the certificates — emptiness of a response correspondence at an information state — not their *cause*, and conflating the two would misdescribe the companions. The dynamic exit certificate is not epistemic: it defeats full-information control. The fibre criterion is not a policy-viability obstruction: it concerns exact certification, a

verdict rather than a control. The certainty-equivalence trap is not necessarily informational: it can occur under an injective observation, because the restriction is on the policy class. Accordingly, the strongest claim the framework supports is: observation-based viability requires the existence of a common recursively safe selector from information states to admissible responses; the obstruction certificates of the companions are checkable sufficient conditions under which no such selector exists; in finite-horizon finite systems the backward recursion characterizes selector existence exactly; in continuous or infinite-horizon settings the certificates are sound local exclusion tests. Within those limits the selector framework is the spine on which the companion results, and the extensions that follow, assemble.

8. Outlook: the stochastic selector

The belief-state safety value of the companion paper’s probabilistic development is the stochastic extension of the recursive selector: Γ_N replaced by the value recursion over distributional information states, emptiness replaced by deficit from unity. Its exact rational theory — piecewise-linear values, support degeneration, and the agreement between certificate boundaries and value boundaries on the audited classes — is developed in the sequel.

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